RT-Thread UART Developer Guide

ID: RK-KF-YF-90

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Security Level: ☐Top-Secret ☐Secret ☐Internal ■Public

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Preface

Overview

Product Version

Chipset	RT-Thread Version
all	lts-v3.1.x/master
all	lts-v4.1.x/master

Intended Audience

This document (this guide) is mainly intended for:

Technical support engineers

Software development engineers

Revision History

Version	Author	Date	Change Description
V1.0.0	Huibin Hong	2019-06-13	Initial version
V1.1.0	Steven Liu	2020-05-15	Updated version
V1.2.0	Huibin Hong	2024-08-15	Support RTT V4.1 or newer

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1. Rockchip UART Features

UART (Universal Asynchronous Receiver/Transmitter)

- Compatible with 16550A
- Some UARTs support auto flow control, please refer to the datasheet
- Support interrupt and DMA mode
- Support baudrate up to 4000000 or 8000000, please refer to the datasheet
- Some SOC support hardware control RS485, please refer to the datasheet

2. Software

2.1 Code path

UART framework:

```
components/drivers/include/drivers/serial.h
components/drivers/serial/serial.c
components/libc/termios/posix_termios.c
components/libc/termios/posix_termios.h
```

UART driver:

```
bsp/rockchip-pisces/drivers/drv_uart.c
bsp/rockchip-pisces/drivers/drv_uart.h
```

UART test code:

```
bsp/rockchip-common/tests/termios_test.c
```

2.2 Configuration

Enable UART, and have /dev/uart0..9.

```
RT-Thread rockchip RKxxxx drivers  --->
    [*] Enable UART
    [*]   Enable UART0
    [ ]   Enable UART1
    [*]   Enable UART2
    [ ]   Enable UART3
    [ ]   Enable UART4
    [ ]   Enable UART5
    [ ]   Enable UART6
    [ ]   Enable UART7
    [ ]   Enable UART8
    [ ]   Enable UART9
```

list_device shows UART devices:

```
msh >list_device
device         type         ref count
------ -------------------- ----------
uart7  Character Device     0
uart6  Character Device     0
uart5  Character Device     0
uart4  Character Device     2
uart3  Character Device     0
uart2  Character Device     0
uart1  Character Device     0
uart0  Character Device     0
```

2.3 UART test

Enable UART test code:

```
lts-v3.1.x:
RT-Thread Components  --->
    Device virtual file system  --->
        [*] Using device virtual file system
        -*- Using devfs for device objects
    POSIX layer and C standard library  --->
        [*] Enable termios feature

RT-Thread bsp test case  --->
    RT-Thread Common Test case  --->
        [*] Enable BSP Common TEST
            [*]   Enable BSP Common UART TEST

lts-v4.1.x or newer:
RT-Thread Components  --->
    [*] DFS: device virtual file system  --->
        [*]   Using devfs for device objects
    C/C++ and POSIX layer  --->
        POSIX (Portable Operating System Interface) layer  --->
           [*] Enable POSIX file system and I/O
               [*]   Enable I/O Multiplexing select() <sys/select.h>
```

```
                [*]   Enable Terminal I/O <termios.h>

RT-Thread bsp test case  --->
    RT-Thread Common Test case  --->
        [*] Enable BSP Common TEST
            [*]   Enable BSP Common UART TEST
```

UART test command:

```
    receive data:
    termtest r /dev/uart1 115200
    send data:
    termtest s /dev/uart1 115200
    receive then send:
    termtest t /dev/uart1 115200
    externel loopback:
    termtest l /dev/uart1 115200
```

2.4 Baud rate

Support all the baud rate under or equal to 1500000, the upper baud rate depands on the SOC's clcok tree.

2.5 Console configuration

```
RT-Thread Kernel  --->
    Kernel Device Object  --->
    [*] Using console for rt_kprintf
        (128) the buffer size for console log printf
        (uart0) the device name for console
```